Calibration Trending System — Decimal Places & Numerical Precision Reference

This document serves as the master specification for all decimal place settings, numerical formatting rules, and significant figures used across the Python backend, desktop UI, PDF reports, and legacy HTBasic factor generators.

1. Standard Resistors (0.1 Ω to 1 GΩ)

Standard resistors use the dynamic **Significant Figures Rule** (`_decimals_for(nominal)` in Python / `getDecimalsForNominal(nominal)` in JavaScript) to ensure consistent relative precision across all 11 decades:

Formula: `Decimals = max(0, 9 - integer_digits)`

Resistance Decade	Nominal Value	Measured Value Decimals	Display Example	Uncertainty (k=2)	95% Acceptance Limits
0.1 Ω	0.1 Ω	**8**	`0.09999894 Ω`	`6.10e-08 Ω`	`[0.09999876 ↔ 0.09999887]`
1 Ω	1 Ω	**8**	`1.00001255 Ω`	`1.20e-07 Ω`	`[1.00001180 ↔ 1.00001320]`
10 Ω	10 Ω	**7**	`9.9999465 Ω`	`2.50e-06 Ω`	`[9.9999420 ↔ 9.9999510]`
100 Ω	100 Ω	**6**	`100.000255 Ω`	`1.50e-05 Ω`	`[100.000210 ↔ 100.000290]`
1 kΩ	1,000 Ω	**5**	`1,000.01137 Ω`	`4.40e-04 Ω`	`[1,000.00693 ↔ 1,000.01710]`
10 kΩ	10,000 Ω	**4**	`10,000.0440 Ω`	`2.00e-03 Ω`	`[10,000.0380 ↔ 10,000.0500]`
100 kΩ	100,000 Ω	**3**	`100,000.687 Ω`	`3.50e-02 Ω`	`[100,000.620 ↔ 100,000.750]`
1 MΩ	1,000,000 Ω	**2**	`1,000,004.20 Ω`	`4.20e-01 Ω`	`[1,000,003.80 ↔ 1,000,004.60]`
10 MΩ	10,000,000 Ω	**1**	`10,000,134.0 Ω`	`3.00e+00 Ω`	`[10,000,128.0 ↔ 10,000,140.0]`
100 MΩ	100,000,000 Ω	**0**	`100,000,700 Ω`	`5.00e+01 Ω`	`[100,000,650 ↔ 100,000,750]`
1 GΩ	1,000,000,000 Ω	**0**	`999,916,000 Ω`	`8.00e+02 Ω`	`[999,910,000 ↔ 999,922,000]`

2. DC Voltage Reference (Fluke 732A / 732C)

Output Terminal	Parameter	Decimals	Format Code	Live Display Example
10 V Output	Measured Voltage	**8**	`%.8f`	`10.00008388 V` *(9 decimals in Excel/DB)*
10 V Output	95% Acceptance Limits	**8**	`%.8f`	`[10.00008012 ↔ 10.00008745]`
1 V Output	Measured Voltage	**8**	`%.8f`	`0.99998244 V`
1 V Output	95% Acceptance Limits	**8**	`%.8f`	`[0.99998015 ↔ 0.99998472]`
Uncertainty (nV)	10 V & 1 V Outputs	**1**	`%.1f`	`35.0 nV / 24.4 nV`
Ambient Pressure	Barometric Reading	**0**	`%d`	`1013 hPa / 845 hPa`

3. AC Voltage Dividers (ET 10mV / ET 100mV)

Divider Model	Parameter	Decimals	Format Code	Live Display Example
ET 10mV (100:1)	Measured Ratio (All 6 Frequencies)	**5**	`%.5f`	`101.07490`
ET 100mV (10:1)	Measured Ratio (All 6 Frequencies)	**5**	`%.5f`	`10.00037`
95% Acceptance Limits	Both Dividers	**5**	`%.5f`	`[101.07247 ↔ 101.07842]`
Uncertainty of Ratio	All 6 Frequencies (1 kHz to 1 MHz)	**1**	`%.1f`	`3.7 ppm , 21.0 ppm , 120.0 ppm`
DIVMEANS.TXT File	10mV Block (Lines 10–27)	**4**	`f"{val:.4f}"`	`100.9039`
DIVMEANS.TXT File	100mV Block (Lines 28–45)	**5**	`f"{val:.5f}"`	`10.04839`

4. Waveform Generator (Keysight 33622A)

Test Group	Parameter	Decimals	Format Code	Live Display Example
Calibration Factors	11 RMS Points (10mV to 3V)	**5**	`%+.5f`	`-0.05902 dB / +0.00120 dB`
Flatness Error	4 Frequencies (2 MHz to 10 MHz)	**3**	`%+.3f`	`-0.030 dB / -0.060 dB`
95% Acceptance Limits	11 cf Series	**5**	`%+.5f`	`[-0.06214 ↔ -0.05590]`
90xxxx_Extended.txt	11 cf Points in File	**5**	`f"{val:+.5f}"`	`-0.05902 ! 10mV, 4MHz dB error`

5. Metrology Analysis & Drift Metrics

Metric	Target Standards	Decimals	Unit	Live Display Example
Relative Error	Resistors, Dividers, 732A	2	ppm	`+10,749.00 ppm / +11.37 ppm`
Relative Error	33622A Flatness	3	dB	`-0.012 dB`
Observed Drift	Resistors, Dividers, 732A	2	ppm	`+0.77 ppm / -2.00 ppm`
Observed Drift	33622A Flatness	4	dB	`-0.0031 dB`
Stability Allowance	Manufacturer Specs	2	ppm	`±5.23 ppm / ±2.50 ppm`
Regression Slope	Annual Drift Rate	3	Scientific	`1.755e-06 Ω/year / 4.210e-04 dB/yr`
Fit Residual (Syx)	Standard Error	3	Scientific	`1.968e-03`

6. Legacy HTBasic Template-Matching Files

The legacy automation files use **Template-Matching (`format_like` in `htbasic_format.py`)** to dynamically match the number conventions of the HP 3458A test programs character-for-character:

- **RESVALUE.TXT**: Plain numbers matching line formatting (**3 to 4 decimals**, e.g. `100000.759`).
- **RESNOM.txt**: Scientific notation coefficient (**6 to 7 decimals** coefficient, e.g. `1.0000134E+7` , `0.999916E+9`).
- **Station Files (286300.txt / 286325.txt)**: Exactly **9 decimals** with no leading zero (e.g. `.100000761`).

7. Where to Customize Decimals in the Codebase

A. Python Resistor Significant Figures

- **Files**: `MainApp.py` and `report_pdf.py`
- **Function**: `_decimals_for(nominal)`

```
def _decimals_for(nominal: float) -> int:
    integer_digits = len(str(abs(int(nominal)))) if abs(nominal) >= 1 else 1
    return max(0, 9 - integer_digits)
```